

Mikey (Equilibrium) — Hackathon MVP Technical Specification

Format: Epics → User Stories with acceptance criteria **Event:** RevenueCat Shipaton · **Platform:** Native Android **Date:** 2026-06-06 · **Status:** Draft for approval

1. Context — why this spec exists

Mikey is *"the first game in the world with an AI coach for martial arts & the gym"* (from the Equilibrium deck). It uses the phone camera + on-device pose detection to **correct technique in real time** ("Hey, fix your torso!"), turns performance into **RPG stats**, and drives progression through belt/championship ladders.

Two things changed from the deck and drive this spec:

1. **Business model.** The deck's flat **premium subscription** is replaced by a **3-tier model** whose centerpiece is a **commitment/stake** option — users put money on the line and earn most of it back by training consistently. This is a behavioral-retention mechanic, not just a paywall.
2. **Scope.** We are building a **hackathon MVP** for Shipaton that must *prove the loop end-to-end in a 3-minute demo*: camera correction → stats → streak/commitment → monetization, wrapped in a non-corny narrative.

What the MVP must prove: a real person can stand in front of an Android phone, get corrected on a **karate** technique and a **gym** rep set, watch real **stats** move, see the **streak/Vow** update, and pass through a **real RevenueCat purchase** + a **real commitment-deposit** flow — all in one sitting.

2. Locked decisions (scope guardrails)

Decision	Choice
MVP training tracks	Karate (technique correction) + Gym reps (rep counting/form)
Platform	Native Android (Kotlin), CameraX + on-device pose
Monetization tiers	Free , Premium (subscription), Commitment Pay (stake)
Premium billing	RevenueCat over Google Play Billing (Shipaton requirement)
Commitment deposit + 2/3 cash-back	External rail (Stripe) as a refundable deposit, kept separate from store IAP
Stats derivation	Real-derive speed/agility/endurance/technique from pose; proxy strength & intelligence
Plot	"The Vow" (Restoration) — diegetic streak
Visual identity	Tactical HUD (live training) + Ink & Dojo accents (menus/progression)
"Mikey"	The AI coach persona (the correcting voice/guide) — <i>confirm if it's actually the brand/avatar</i>

3. Training, Progression & Narrative (revised)

Locked-decision delta		
Topic	Old (deck/spec)	Revised
Source of "cool"	narrative wrapper / hero myth	the produced training loop (escalation), not narration
Opening	Prologue: lore, then make the Vow first	animated cold-open showing other players (PvP seed) → straight into Level 0
Narration	Mikey monologue, motivational	two voices — coach = functional cues; frame = optional 3rd-person fable, never 2nd-person hype

Production	unspecified	animated, not filmed ; slow-mo / haptic / HUD flare on lock-in
First content	Heian base kata	L0 combine (max-rep tests) → L1 zuki chudan
Belt ceiling	belt → championship ladder	honest auto-grade caps at ~brown ; black belt → go to a real dojo (post-MVP)
Gym ↔ karate	separate tracks	soft gate : technique tiers need conditioning built in gym mode
Vow/stake timing	onboarding, before play	after L0 + L1 aha

3.1 Design doctrine

- **Cool comes from the loop, produced — not from words.** "Do → correct → faster," shot with escalation, is a training montage; montages work because demand rises. No narration is required to make it cool.
- **Cringe rule:** never address the player's readiness or promise transformation ("are you ready to transform" is the cringe vector). The app observes and instructs; it does not pitch.
- **Two voices, kept separate.** *Coach* (in-session) = functional single-word cues only ("lower," "again"). *Frame* (between sessions) = optional, quiet, third-person fable in a Shadow-Fight register — mythic and economical, never motivational.
- **Realism register: High Kick Girl, not Ip Man.** Real, grounded, heightened in presentation — never wire-fu fantasy. (Forced by the product anyway: the user performs real reps on a phone cam.)

3.2 Narrative frame (minimal)

- Theme retained — technique is being lost and diluted (mcdojos, clout) — but delivered as a quiet, optional third-person fable between sessions. **The streak is the story:** show it via the form-restoration meter and stat sheet; don't narrate it.
- **Honesty is the spine.** The app grades you to ~brown belt, then *sends you to a real dojo*. That's the literal opposite of an mcdojo — it refuses to sell a fake belt — so the anti-dilution theme

becomes something the product *does*, not says. (Also seeds a post-MVP dojo-referral business — see §8.)

3.3 The session loop

do → correct → repeat → faster. **Each round adds exactly one demand** (one more scored joint/axis), so difficulty ramp and scoring are the same system. "Faster" = tempo, genuinely measured by the engine. Production beats: animated master demo → live HUD skeleton → single-word cues → jade flare + hard haptic + sound on each clean rep → accelerating cadence pulse in the speed round → capstone rep in slow-mo with the heaviest feedback, visible form-fragment restoration, and a live stat tick.

3.4 Level structure (MVP)

Level 0 — Combine (measurement). Max push-ups, max squats (+ optional). Form-gated: only clean reps count. Camera-safe set only (push-ups, squats reliable; floor work noisier). Populates **real** stat baselines — upgrades Strength/Endurance from "proxy" to measured (see §9) — and doubles as the safest demo open, since rep-counting is the engine's most reliable output.

Level 1 — First technique: zuki chudan. Short animated demo → "Practice required — start now?" → the §3.3 loop. Chosen as a canonical first karate fundamental that is also beginner-legible and the most camera-reliable strike (frontal).

3.5 Progression — belt-tiered rubrics

One technique = one scorer with belt-tiered rubrics. Each rank adds scored dimensions and tightens thresholds on the *same* landmarks:

- Yellow: rough shape + chudan height
 - hikite + guard
 - stance
 - hip drive + tempo
 - **kime** (sharp acceleration to full extension, then abrupt stop — measurable from the velocity profile)

Honest ceiling: a phone cam can grade shape, angles, tempo, and a kime *proxy*; it cannot grade real power, breathing/kiai, timing, or "feeling." Auto-grading therefore caps at **~brown belt**. **Black belt = the app directs you to a real dojo** (post-MVP). Certifying dan ranks is explicitly out of scope.

3.6 Cross-track gating (karate ↔ gym)

Soft gate, never a hard lock. Higher technique tiers require performance you can't physically produce without conditioning (e.g., black-belt-grade kime needs explosive power built in gym mode). The requirement emerges organically from the scoring thresholds — no "LEVEL LOCKED — train strength" wall.

3.7 Monetization timing

The Vow/commitment stake is offered **after** Level 0 + Level 1 (the aha), not at onboarding. Rationale: deliver felt value before any ask — a Vow to protect a form you've already restored is real; one sworn before you've touched the product is hollow. (Commitment model and billing split: see §8 revision.)

User-story impact

- Supersedes **US-1.1** (no Vow-first prologue); reframes **US-1.2** timing (post-aha).
- Rewrites **US-3.1/3.2/3.3** around the one-demand-per-round ramp. **US-3.4** (kata sequence) deferred — the first technique is a single strike, not a sequence.
- Adds **L0 combine** stories that feed **US-5.1** (real baselines, not proxy).
- **US-8.1** belt ladder capped at brown + dojo-handoff as the terminal rung.

This is §3 only. **§8 (commitment decomposition + RevenueCat/Stripe billing split)** and **§9 (stats, now with L0 measured baselines)** also changed materially — want me to produce those two next?

4. Art direction (with references)

Dominant: Tactical HUD / Telemetry for the live training screen — a glowing pose-skeleton overlaid on the live camera, correction cues as crisp callouts, stats as live telemetry.

- **Refs:** *Sifu* combat UI · *Ghost of Tsushima* duel HUD · WHOOP/Strava data-as-hero · *Death Stranding* HUD.
- **Palette:** near-black #0B0E11 base, bone text #EDE6D8, one **hot accent** — ember #FF5A1F (error/correction) + jade #2BD9A8 (correct/locked-in). Belt colors reserved for progression.
- **Type:** condensed grotesque for HUD numerals (e.g., Rajdhani / Saira), humanist sans for body.
- **Motion:** skeleton joints pulse jade when correct, ember when off; correction callouts slide in for ~1.2s; haptic tick on lock-in.

Accent: Ink & Dojo / Sumi-e for menus, the Vow screen, chapter/story beats, and the stat sheet — brush strokes, washi-paper texture, vermilion seal #C8472E, belt-color accents, restrained and premium.

- **Refs:** *Ghost of Tsushima* menus · *Okami* · *Sifu* posters · Muji / Teenage Engineering restraint · Headspace warmth.

Why this combo: HUD gives the "the AI is really correcting me" credibility (and is cheap to render — overlays on the camera feed); Ink & Dojo gives the premium, cultural, *un-comfy* frame for the money/commitment/story surfaces where trust matters. nigga

5. Product scope (MVP)

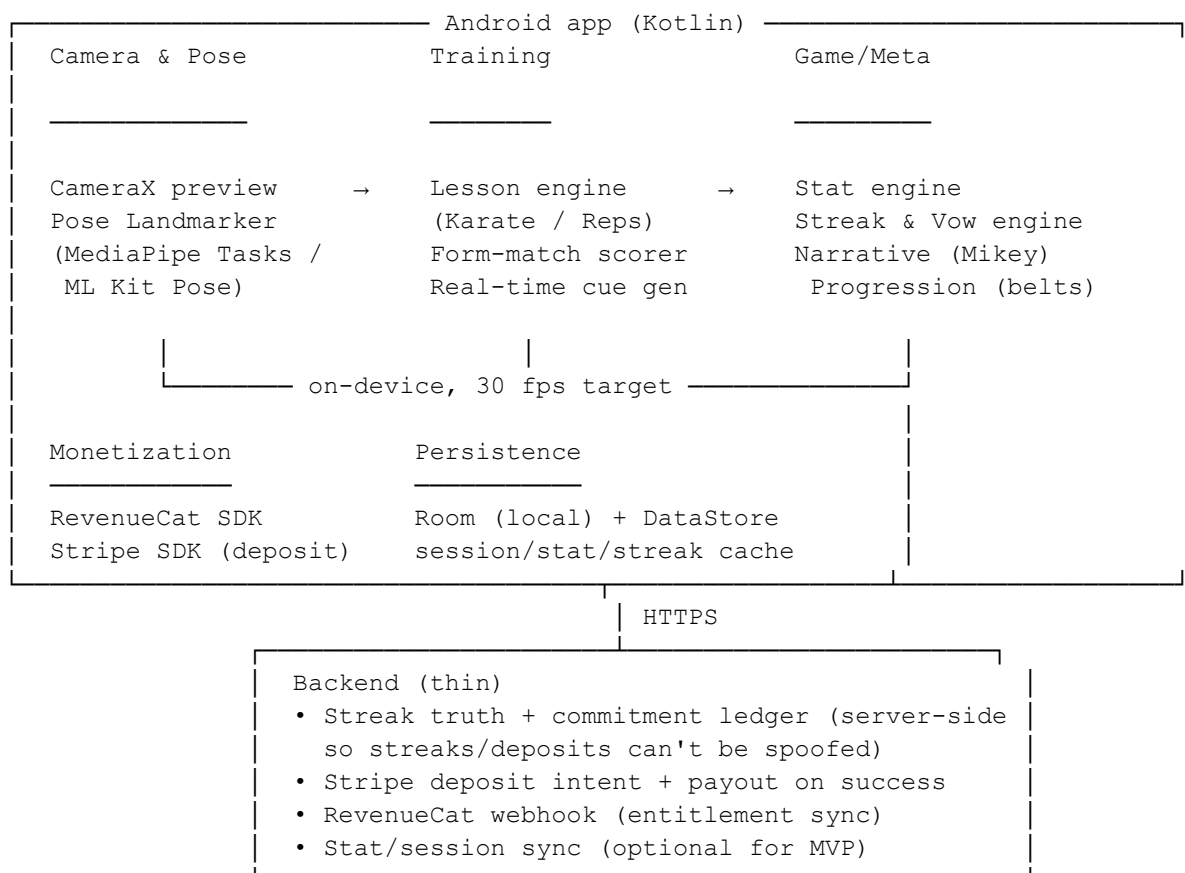
IN:

- Onboarding → plan selection → (optional) Vow setup.
- **Karate lesson mode:** 1 lesson, 3–4 elements (e.g., front stance *zenkutsu-dachi*, mid block, front kick *mae-geri*, a short kata sequence) with **real-time correction**.
- **Gym reps mode:** 2 exercises (squat, push-up) with **rep counting + depth/form correction**.
- **Real-time pose engine** (on-device).
- **Stat sheet** (5 stats) updated from session data.
- **Streak + Vow engine** with diegetic form-decay visual.
- **3-tier monetization:** Free / Premium (RevenueCat) / Commitment Pay (Stripe deposit + payout).
- **Session summary** + minimal history.
- **Belt/championship ladder** visualized (first rung interactive).

OUT (post-MVP): multiplayer, tournaments, MR/AR, B2B2C gym portal, full discipline catalog (boxing/muay thai/etc.), social, leaderboards, cloud avatar rendering.

6. System architecture

Native Android, organized into bounded contexts (DDD per project conventions), thin backend for money + streak truth.



Pose pipeline (real-time loop): CameraX frame → Pose Landmarker (33 landmarks) → normalize (scale/orient invariant) → per-mode scorer → cue + score → HUD render + stat accumulators. Target ≥20–30 fps, correction latency < ~200 ms.

Scoring approach:

- **Karate:** compare normalized joint **angles** (key joints per element) against a **reference pose/sequence**; flag the worst-offending joint(s) → one prioritized cue at a time ("Hips lower"). Sequence (kata) = ordered keyframe matching with tempo tolerance.
- **Reps:** track a primary joint-angle signal (knee for squat, elbow for push-up) through a **state machine** (top→bottom→top = 1 rep); gate "good rep" on depth + back/torso alignment.

Key tech choices: Kotlin, CameraX, **MediaPipe Tasks Pose Landmarker** (or ML Kit Pose) on-device, Room + DataStore, RevenueCat Android SDK, Stripe Android SDK, lightweight backend (any — Node/Supabase/Firebase Functions) for streak truth + Stripe.

Why server-side streak/ledger: money rides on the streak, so the **authoritative streak counter and the commitment ledger live on the backend** (client reports completed sessions; server validates and advances). Prevents trivial clock/spoof cheating during a money flow.

7. Data model (key entities)

- **User** {id, displayName, plan(FREE|PREMIUM|COMMITMENT), createdAt}
- **Session** {id, userId, mode(KARATE|REPS), startedAt, durationS, elements[], techniqueScore, reps?, qualifies:bool}
- **StatSheet** {userId, strength, speed, agility, endurance, intelligence, updatedAt} (each 0–100, EWMA-smoothed)
- **Streak** {userId, current, longest, lastQualifiedDate, status}
- **Vow / Commitment** {userId, stakeAmount, payoutAmount(2/3), termDays(30), startDate, status(ACTIVE|WON|FORFEITED), depositRef, payoutRef}
- **Form (narrative)** {id, name, restorationPct, decayRatePerMissedDay} — drives the diegetic decay visual.
- **Progression** {userId, discipline, beltRank, ladderRung}

8. Monetization & commitment engine

Three tiers. **Premium subscription must flow through RevenueCat/Play billing** (Shipaton). The **Commitment deposit is a refundable real-money deposit on an external rail (Stripe)** — deliberately *not* an IAP, because it's a refundable deposit for a real-world coaching commitment, not a consumable digital good.

Tier	Cost	Gets	Mechanic
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Free	\$0	1 karate lesson + reps, basic correction, basic stat sheet	Entry
Premium	P (~\$5/mo)	All lessons, detailed correction, full tracking/history	RevenueCat subscription
Commitment Pay	stake $\approx 3P$ (~\$15)	Premium features for the term + skin in the game	Stripe deposit; 30-day streak → 2/3 ($\approx \\$10$) paid back as real cash via Stripe payout; net $\approx P$

Streak gift (all tiers): reaching a **30-day streak** grants **free Premium time** (e.g., +1 month).

Commitment math (parameterized; example $P=\$5$): stake \$15 → success refunds \$10 → **net \$5**; failure forfeits the refund (you paid \$15 and keep Premium access for the term). Numbers are config, not hardcoded.

Refund/payout flow: deposit captured at Vow start → backend tracks the 30-day window against authoritative streak → on **WON**, trigger Stripe payout of `payoutAmount`; on **FORFEITED**, no payout. Streak gift applied independently on any 30-day streak.

⚠️ **Risk flags (carry into build):**

- App-store policy: keep the Stripe deposit **clearly separate** from in-app digital purchases; frame as a real-world fitness-commitment deposit. Validate against Play policy before public release (acceptable for hackathon demo).
- Real payouts need a funded Stripe account + payout rails; for the demo, use **Stripe test mode** end-to-end and show the payout call succeeding.

9. Stat derivation

Stat	Source	Heuristic (MVP)
Speed	Real	Peak joint angular velocity on strikes/reps (normalized)
Agility	Real	Stance-transition time + range of motion between keyframes
Endurance	Real	Reps/tempo sustained before form-quality decay within a session
Technique	Real	Pose-match accuracy vs reference → also drives belt progression
Strength	Proxy	Work proxy = tempo × range × isometric hold-time (e.g., squat depth-hold)

Stats are **EWMA-smoothed** across sessions (no wild single-session swings), each rendered 0–100 on the Ink & Dojo stat sheet.

10. Technical specification — user stories

Convention: **AC** = acceptance criteria (Given/When/Then). Stories sized for a hackathon; ★ = demo-critical (must work live).

Epic 1 — Onboarding & plan selection

- **US-1.1** ★ As a new user, I want a 3-screen onboarding (the problem → Mikey's promise → make your Vow) so that I understand the product fast.
 - AC: Given first launch, when I open the app, then I see ≤3 intro screens in Ink & Dojo style and land on plan selection.
- **US-1.2** ★ As a user, I want to choose **Free / Premium / Commitment Pay** with clear costs and the 2/3-back math so that I can pick consciously.
 - AC: Each tier shows price, benefits, and (for Commitment) the stake/refund/net example; selecting routes to the correct purchase flow.
- **US-1.3** As a user, I want camera-permission priming before the system dialog so that I grant access willingly.
 - AC: Rationale screen precedes the OS permission prompt; denial routes to a graceful "camera required" state.

Epic 2 — Camera & pose engine

- **US-2.1** ★ As a user, I want the app to detect my body from the front camera in real time so that it can correct me.
 - AC: With me fully in frame, a 33-landmark skeleton tracks at ≥20 fps; partial/no-body shows a "step back / get in frame" guide.
- **US-2.2** ★ As a user, I want pose tracking to be position/scale tolerant so that it works in a normal living room.
 - AC: Detection holds across a reasonable distance range and lighting; normalization makes scoring invariant to my distance from the phone.
- **US-2.3** As a user, I want an on-screen framing helper (silhouette/target box) so that I place my phone correctly.
 - AC: A framing overlay confirms (jade) when my full body is captured and angle is acceptable.

Epic 3 — Karate lesson mode (real-time correction)

- **US-3.1** ★ As a learner, I want Mikey to demo a karate element, then watch me attempt it so that I learn by doing.
 - AC: Each element shows a reference demo, then a "your turn" capture with the live HUD skeleton.

- **US-3.2** ★ As a learner, I want **one prioritized correction cue at a time** ("Hips lower", "Fix your torso") so that I'm not overwhelmed.
 - AC: The worst-offending joint(s) drive a single cue + ember highlight; on correction it flips jade with a haptic tick.
- **US-3.3** ★ As a learner, I want a **technique score (0–100)** per element so that I know how clean I was.
 - AC: Score derives from normalized joint-angle deviation vs reference; shown at element end; feeds Technique stat + belt progress.
- **US-3.4** As a learner, I want a short kata sequence scored on order + tempo so that I practice flow, not just static poses.
 - AC: Ordered keyframe match with tempo tolerance; missed/out-of-order keyframes are flagged.

Epic 4 — Gym reps mode (counting + form)

- **US-4.1** ★ As a user, I want accurate **rep counting** for squats and push-ups so that I trust the tracking.
 - AC: A top→bottom→top state machine counts a rep only on full range; partial reps are not counted (and flagged).
- **US-4.2** ★ As a user, I want **form correction during reps** (squat depth, back/torso line; push-up hip sag) so that I train safely.
 - AC: Below-depth or misaligned reps trigger a real-time cue and count as "no-rep" or "shallow"; good reps confirm jade.
- **US-4.3** As a user, I want a set summary (reps, good-rep %, tempo) so that I see quality, not just quantity.

Epic 5 — Stat sheet (RPG stats)

- **US-5.1** ★ As a player, I want a **5-stat sheet** (strength/speed/agility/endurance/intelligence) updated from my sessions so that progress feels RPG-real.
 - AC: After a qualifying session, affected stats move per §9 heuristics, EWMA-smoothed, on the Ink & Dojo sheet.
- **US-5.2** As a player, I want each stat to show *why* it moved (e.g., "Speed +3 from strike velocity") so that the numbers feel earned, not random.
- **US-5.3** As a player, I want a small "intelligence" micro-quiz after a lesson so that coachability is measured, not faked.

Epic 6 — Streak & Commitment ("The Vow")

- **US-6.1** ★ As a committed user, I want a **streak counter** that advances when I complete a qualifying session that day so that consistency is visible.
 - AC: A day counts when ≥1 qualifying session is logged (server-validated); the counter and "longest" update; local-midnight day boundary for MVP.
- **US-6.2** ★ As a user, I want the **diegetic decay**: a missed day visibly erodes the restored "form" so that the plot and my consistency are one thing.
 - AC: On a missed day, the form-restoration meter drops by its decay rate and Mikey notes it; consecutive clean days restore it.
- **US-6.3** ★ As a Commitment-Pay user, I want a **Vow contract** (stake, term=30, 2/3 payback, start date) created at purchase so that the terms are explicit.
 - AC: Vow entity created with Stripe deposit ref; status ACTIVE; terms shown on a sealed Ink & Dojo "Vow" screen.

- **US-6.4** ★ As a Commitment-Pay user, I want to **get my 2/3 back automatically when I complete the 30-day streak** so that the incentive is real.
 - AC: On authoritative streak == term, status→WON, backend triggers Stripe payout of `payoutAmount`; UI confirms cash-back. On break, status→FORFEITED, no payout.
- **US-6.5** As any user, I want a **30-day streak gift** (free Premium time) so that consistency is rewarded even without a stake.

Epic 7 — Monetization (RevenueCat + deposit)

- **US-7.1** ★ As a user, I want to **buy Premium via the store** so that purchasing is trusted and Shipaton-compliant.
 - AC: RevenueCat offering shown; successful purchase grants the `premium` entitlement; entitlement gates premium features.
- **US-7.2** ★ As a Commitment-Pay user, I want to **place my stake deposit via Stripe** so that the commitment is backed by real money.
 - AC: Stripe payment sheet captures the deposit (test mode for demo); success creates the Vow and grants premium features for the term.
- **US-7.3** As a user, I want entitlements to persist/restore across launches so that I keep what I paid for.
 - AC: RevenueCat entitlement + backend ledger restore on login/reinstall.

Epic 8 — Progression & narrative (Mikey)

- **US-8.1** ★ As a player, I want **belt/ladder progression** tied to technique scores so that improvement has a destination (per deck).
 - AC: Reaching technique thresholds advances belt/ladder rung; the Dojo→World ladder is visualized; first rung interactive in MVP.
- **US-8.2** As a player, I want **Mikey's chapter beats** (weekly, calm mentor tone) so that the story rewards consistency.
 - AC: Chapter 1/Day 1 narration plays in MVP; tone is terse/respectful, never corny.

Epic 9 — Session summary & history

- **US-9.1** ★ As a user, after a session I want a summary (technique score, reps, stat deltas, streak/decay, Mikey line) so that the loop closes satisfyingly.
- **US-9.2** As a user, I want a minimal history list so that I can see past sessions.

Non-functional requirements

- **Performance:** pose loop ≥20 fps mid-range Android; correction latency <~200 ms; on-device (no frame upload).
 - **Privacy:** camera frames processed **on-device**, never uploaded; explicit camera consent; stats/streak only sync as data, not video.
 - **Reliability:** graceful degradation if pose confidence drops (pause + reframe guide); offline-tolerant session capture, sync streak when online.
 - **Security:** streak + commitment ledger server-authoritative; Stripe/RevenueCat secrets server-side; no secrets in the APK.
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11. Demo script (Shipaton, ~3 min)

1. Onboarding → pick **Commitment Pay**, place **Stripe test deposit** → Vow sealed. (US-1.2, 6.3, 7.2)
 2. **Karate**: attempt front stance → Mikey: "Hips lower" (ember→jade) → technique score. (US-3.2, 3.3)
 3. **Reps**: 3 squats — one shallow no-rep, two clean. (US-4.1, 4.2)
 4. **Stat sheet** animates: Speed/Technique/Strength move, with reasons. (US-5.1, 5.2)
 5. **Streak** → Day 1 of Vow; show the decay concept (what a miss would do). (US-6.1, 6.2)
 6. Fast-forward/simulate Day 30 → **Stripe payout** of 2/3 fires → "your \$10 is back." (US-6.4)
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12. Verification (how we'll test end-to-end)

- **Pose/scoring**: record reference attempts; assert technique score rises with cleaner form and rep state machine counts correctly vs hand counts; test framing edge cases (out-of-frame, low light).
 - **Stats**: unit-test each heuristic with synthetic landmark streams; assert EWMA smoothing and "reason" attribution.
 - **Streak/Vow**: simulate day advancement on the backend; assert streak advances on qualifying sessions, breaks on a miss, and WON triggers exactly one payout; FORFEITED triggers none.
 - **Money**: RevenueCat sandbox purchase grants entitlement; Stripe test-mode deposit + payout round-trip succeeds and is reflected in UI.
 - **Demo dry-run**: execute §11 on a physical mid-range Android device, end-to-end, twice.
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13. Open assumptions to confirm

- "Mikey" = AI coach persona (assumed). If it's the brand/player avatar, the coach needs a separate name.
 - Exact P (premium price), stake multiple ($\approx 3\times$), and payback fraction ($2/3$) are config — using \$5 / \$15 / \$10 as the demo example.
 - Backend stack is open (Supabase/Firebase/Node all fine) — chosen at implementation for fastest path to streak-truth + Stripe.
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14. Out of scope (post-MVP)

Multiplayer, tournaments, MR/AR sparring, full discipline catalog (boxing, muay thai, taekwondo, JKD, etc.), B2B2C gym portal, social/leaderboards, cloud-rendered legend avatars, in-house hardware.

